

Lecture 1

Statistical Learning

1.1

Contents

1	What is Statistical Learning?	1
2	Examples of Statistical Learning Problems	2
3	The Supervised Learning Problem	2
4	Goals	2
5	Philosophy	3
6	Unsupervised learning	3
7	Some Examples	4
7.1	Basic Parametric and Non-Parametric Solutions to Supervised Problems	6
8	A Brief History of Statistical Learning	11
9	More on Statistical Learning Goals	11
10	Estimation of $f(\cdot)$	13
11	The Trade-Off Between Prediction Accuracy and Model Interpretability	16

Some figures in this presentation are taken from 'An Introduction to Statistical Learning, with applications in R' (Springer, 2013) with permission from the authors: G. James, D. Witten, T. Hastie and R. Tibshirani.

Most of the material is extracted from the slides of the MOOC course by Hastie and Tibshirani, available at <https://www.r-bloggers.com/in-depth-introduction-to-machine-learning-in-15-hours-of-expert-videos/>.

1.2

1 What is Statistical Learning?

It refers to a vast set of tools/methods/techniques for analyzing and *understanding* data.

It involves:

- building statistical models aimed to predict or estimate *an output variable, or outcome, or response* basing on one or more explanatory variables or predictors (*supervised learning*);
- learning relationships and structures that exist in a set of variables or subjects, when an output variable does not exist (*unsupervised learning*).

1.3

2 Examples of Statistical Learning Problems

See, also, T. Hastie, R. Tibshirani and J. Friedman 'The Elements of Statistical Learning' (Springer, 2008).

- Establish the relationship between salary and demographic variables in population survey data.
- Predict the price of a stock in 6 months from now, on the basis of company performance measures and economic data.
- Customize emails from customers of bank or insurance companies as expressing positive or negative opinions.
- Customize an email spam detection system.
- Classify a tissue sample into one of several cancer classes, based on a gene expression profile.
- Predict whether a patient, hospitalized due to a heart attack, will have a second heart attack. The prediction is to be based on demographic, diet and clinical measurements for that patient.
- Identify the numbers in a handwritten ZIP code, from a digitized image.
- Identify the risk factors for prostate cancer, based on clinical and demographic variables.

1.4

3 The Supervised Learning Problem

Starting point:

- Outcome measurement Y (also called dependent variable, response, target).
- Vector of p predictor measurements X (also called inputs, regressors, covariates, features, independent or explanatory variables).
- In the *regression problem*, Y is quantitative (e.g price, consumption, blood pressure).
- In the *classification problem*, Y is categorical (usually unordered) (employed/unemployed, survived/died, digit 0-9, cancer class of tissue sample).
- We have *training data* $(x_1, y_1), \dots, (x_N, y_N)$. These are observations (examples, instances) of these measurements.

1.5

4 Goals

On the basis of the training data we would like to:

- *Accurately predict unseen test cases.*
- *Understand which inputs affect the outcome, and how.*
- *Assess the quality of our predictions and inferences.*

1.6

Regression vs Classification

Regression:

the outcome is *quantitative*

Classification:

the outcome is *categorical*

Also in this case the taxonomy could not be definitive.

Logistic regression models applied to 'churn' analysis try to explain/predict an outcome variable Y with the following *categories*

- 0 will not churn
- 1 will churn

One of result of logistic regression is the probability

$$P(Y = 1|X_1, X_2, \dots, X_k),$$

that has values in the interval (0,1); we are in presence of a *metric* outcome, that is a quantitative variable.

1.7

5 Philosophy

- It is important to fully understand the ideas behind the various techniques, in order to know how and when to use them.
- Understand the simpler methods first, and then attach with the more sophisticated ones.
- Accurately assess the performance of a method, to know how well or how badly it is working.
Simpler methods often perform as well as fancier ones!
- Statistical learning has important applications in science, industry, finance, management and marketing.
- Statistical learning is a fundamental ingredient in the training of a modern *data scientist*.

1.8

6 Unsupervised learning

- No outcome variable, just a set of predictors measured on sample data.
- Objective is more vague – find groups of subjects that have a similar behaviour, find features that behave similarly, find linear combinations of features that best represent initial features.
- It is not simple to know how well your are doing.
- It is different from supervised learning, but can sometimes be useful also as a pre-processing step preliminary to supervised learning.

1.9

Examples of supervised and unsupervised methods

supervised learning methods:

a response variable is present

- linear regression
- logistic regression
- GAM (generalized additive models)
- boosting
- support vector machines

unsupervised learning methods:

a response variable is not available

- principal components
- factor analysis
- cluster analysis
- multidimensional scaling

1.10

Sometimes the taxonomy is not so neat.

In order to analyze complaints by customers of an insurance company we can consider the text mining classification of the log data from a call center or of the emails received by the company.

Classification can be performed as an unsupervised analysis.

We are in presence of a semi-supervised analysis if a reasoned classification of a reduced training set is made, that generates a 'response variable' (see e.g. the methods used by 'Voices from the blogs').

1.11

7 Some Examples

Example 1 (Analysis of the Belgium wage data set, see Verbeek, 2010, A Guide to Modern Econometrics, John Wiley)). 1472 subjects from the 1994 Belgium wave, source: the European Community Household Panel.

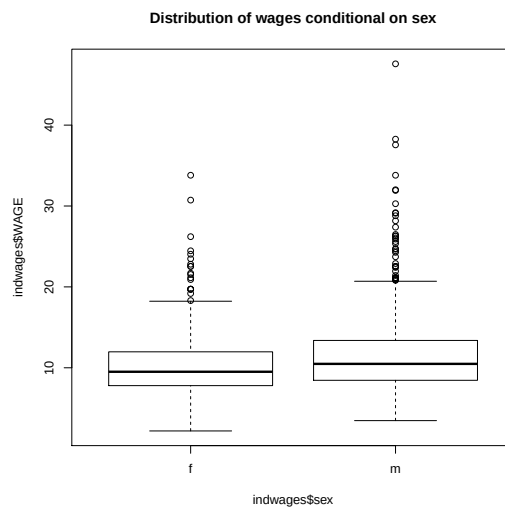
- wage: gross hourly wage rate in euro
- lnwage: $\ln(\text{wage})$
- educ: education level from 1 [low] to 5 [high]
- exper: years of experience
- lnexper: $\ln(1+\text{exper})$
- lneduc: $\ln(\text{educ})$
- male: dummy, 1 if male

We propose a short exploratory analysis of the wage, studied as a function of the sex and the education level.

1.12

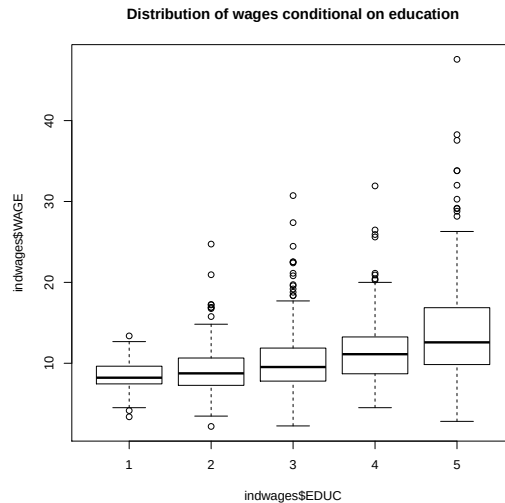
```
indwages <- read.table("bwages.dat", header = T)
indwages <- transform(indwages, sex = factor(MALE, levels = 0:1,
labels = c("f", "m")))

boxplot(indwages$WAGE ~ indwages$sex,
main = "Distribution of wages conditional on sex")
```



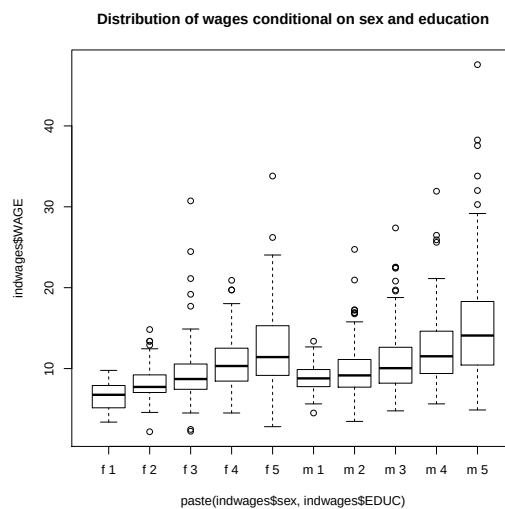
1.13

```
boxplot(indwages$WAGE ~ indwages$EDUC,
        main = "Distribution of wages conditional on education")
```



1.14

```
boxplot(indwages$WAGE ~ paste(indwages$sex, indwages$EDUC),
        main = "Distribution of wages conditional on sex and education")
```



1.15

Problems

- Is the analysis supervised or unsupervised?
The correct answer is: *supervised*
- Build a multiple linear regression model to study the relationship between Y = 'wage' and the explanatory variables X_1 = 'sex' and X_2 = 'education level'.
- Complete the regression model with the variable X_3 = 'experience'.

1.16

7.1 Basic Parametric and Non-Parametric Solutions to Supervised Problems

Some basic parametric and non-parametric solutions are briefly illustrated in order to describe/predict a *metric* outcome Y as function of two predictors X_1 and X_2 .

We assume that a training data set $(y_1, x_{11}, x_{12}), \dots, (y_n, x_{n1}, x_{n2})$ is available.

1.17

Quadratic loss function (OLS), parametric model

The solution to the *parametric* model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + E,$$

where Y , X_1 , X_2 and E are statistical variables referred to observations and residuals, according to the quadratic loss function, is

$$\begin{aligned} \hat{\beta} &= \arg \min_{\beta} \left\{ \frac{1}{n} \sum_{i=1}^n e_i^2 \right\} = \arg \min_{\beta} \left\{ \frac{1}{n} \sum_{i=1}^n [y_i - (\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})]^2 \right\} = \\ &= \arg \min_{\beta} M[Y - (\beta_0 + \beta_1 X_1 + \beta_2 X_2)]^2 = \arg \min_{\beta} M[E]^2 \end{aligned}$$

that corresponds to the well-known OLS estimator

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}.$$

1.18

Quadratic loss function (OLS), non-parametric model

The solution, according to the quadratic loss function, to the *non-parametric* model

$$Y = f(X_1, X_2) + E,$$

where $f(\cdot)$ is a real-valued function, is

$$\hat{f}(x_1, x_2) = \arg \min_f M[Y - f(X_1, X_2)]^2 = M[Y|X_1 = x_1, X_2 = x_2].$$

It is the so-called *regression function*, that links $M[Y|X_1 = x_1, X_2 = x_2]$ to each pair (x_1, x_2) ; it summarizes Y with its average conditional on $(X_1 = x_1, X_2 = x_2)$.

1.19

Absolute value loss function, non-parametric model

The solution, according to the absolute value loss function, to the *non-parametric* model

$$Y = f(X_1, X_2) + E,$$

where $f(\cdot)$ is a real-valued function, is

$$\hat{f}(x_1, x_2) = \arg \min_f M[|Y - f(X_1, X_2)|] = y_{0.5} |_{(X_1=x_1, X_2=x_2)}.$$

It is a variant of the regression function, that links $y_{0.5} |_{(X_1=x_1, X_2=x_2)}$ to each pair (x_1, x_2) ; it summarizes Y with its median (the 50th percentile) conditional on $(X_1 = x_1, X_2 = x_2)$.

1.20

Absolute value loss function, parametric model

The solution, according to the absolute value loss function, to the *parametric* model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + E$$

is the so-called quantile regression estimator

$$\hat{\beta} = \arg \min_{\beta} M[|Y - (\beta_0 + \beta_1 X_1 + \beta_2 X_2)|] = \arg \min_{\beta} M[|E|].$$

See Koenker R 2012 Quantile Regression in R: a vignette. 1-20, <http://cran.r-project.org/web/packages/quantreg/vignettes/rq.pdf>.

1.21

Example 2 (Classification Problem (1)). Figure extracted from James, Witten, Hastie, Tibshirani (2013) *An Introduction to Statistical Learning*, Springer

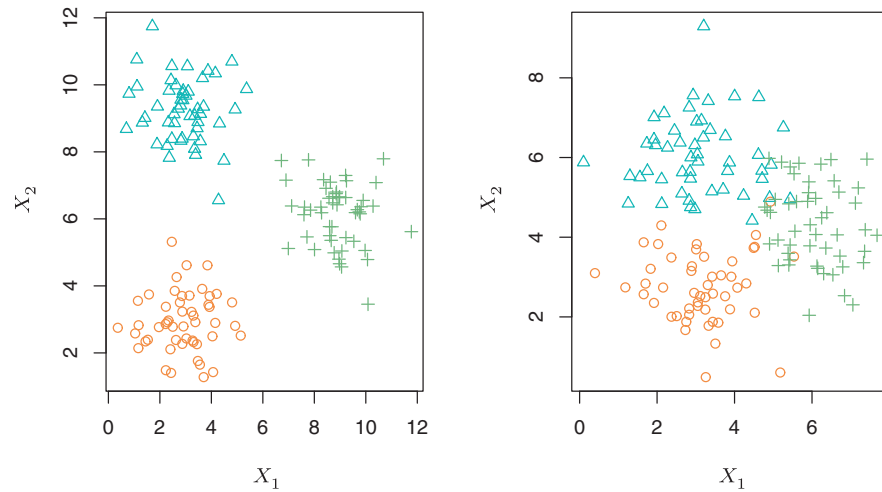


FIGURE 2.8. A clustering data set involving three groups. Each group is shown using a different colored symbol. Left: The three groups are well-separated. In this setting, a clustering approach should successfully identify the three groups. Right: There is some overlap among the groups. Now the clustering task is more challenging.

- Is the analysis supervised or unsupervised?
The correct answer is: *unsupervised*

Example 3 (Classification Problem (2)). Suppose now that the classification of the points in the following Figure 1 is known.

(code to produce the figure)

```
library(mvtnorm)
# n is the number of units in each group
n <- 100
# set.seed sets the random number seed (for reproducibility)
set.seed(123456)
# generation of coordinates in the first group (values of
# variables X1 and X2)
data1 <- rmvnorm(n, mean = c(1, 2), sigma = matrix(c(1, 0, 0,
1), 2, 2))
# assignment of row names
rownames(data1) <- paste("A", 1:n, sep = "")
# generation of coordinates in the second group (values of
# variables X1 and X2)
data2 <- rmvnorm(n, mean = c(3, 1), sigma = matrix(c(1, 0.5,
0.5, 1), 2, 2))
# assignment of row names
rownames(data2) <- paste("B", 1:n, sep = "")
# merge in a unique data set
data <- rbind(data1, data2)
# assignment of variable names
colnames(data) <- c("x1", "x2")
# assignment of outcome variable with category A for the
# points in the first group and B for the points in the
# second group
outcome <- rep(c("A", "B"), each = n)
```

1.23

```
# graph initialization
plot(data, type = "n", xlim = c(-2, 6), ylim = c(-2, 5))
# plot points belonging to the first group
points(data[1:n, ], pch = 16, col = "blue")
# plot points belonging to the second group
points(data[1:n + n, ], pch = 1, col = "red")
# plot a grid as cartesian product between the two sets
# {-2, -1.9, ..., 5.9, 6} and {-2, -1.9, ..., 4.9, 5}
set <- expand.grid(-20:60/10, -20:50/10)
# plot points that are on the grid
points(set, pch = 1, col = "grey", cex = 0.2)
```

1.24

We want to classify some new points (all the points on the grey grid), or, equivalently, we want to define a function to separate the two (known) groups of data.

1.25

To this aim we can use the *k-nearest-neighbours* (KNN) method

- given the coordinates of the point we want to classify, one observe to which group belong the k points in the training data set, nearest to the point we want classify;
- if the proportion of points classified as A is higher than 0.5 then the point, we want to classify, is assigned to group A, otherwise to group B

In order to implement the KNN algorithm one can use the function `knn` available in the package `class`.

Figures 2 and 3 show the classification results with k equal to 3 and 5, respectively.

1.26

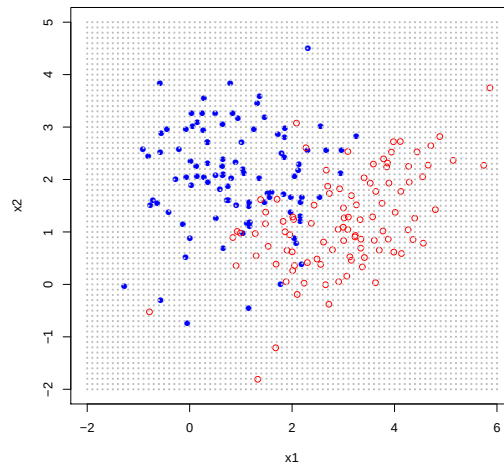


Figure 1: Example of data belonging to two groups. Which group each subject belongs to is known a priori.

```
library(class)
# classification with the KNN algorithm (k=3); result is an
# array with predictions of which group each new subject on
# the grid (set) belongs to
knn.pred <- knn(data, set, outcome, k = 3)
# create a new figure with training set (data) points
plot(data, type = "n", xlim = c(-2, 6), ylim = c(-2, 5))
points(data[1:n, ], pch = 16, col = "blue")
points(data[1:n + n, ], pch = 1, col = "red")
# plot grid points classified in the first group
points(set[knn.pred == "A", ], pch = 16, col = "blue", cex = 0.2)
# plot grid points classified in the second group
points(set[knn.pred == "B", ], pch = 1, col = "red", cex = 0.2)
```

```
# classification with the KNN algorithm (k=5)
knn.pred <- knn(data, set, outcome, k = 5)
# create a new figure with training set (data) points
plot(data, type = "n", xlim = c(-2, 6), ylim = c(-2, 5))
points(data[1:n, ], pch = 16, col = "blue")
points(data[1:n + n, ], pch = 1, col = "red")
# plot grid points classified in the first group
points(set[knn.pred == "A", ], pch = 16, col = "blue", cex = 0.2)
# plot grid points classified in the second group
points(set[knn.pred == "B", ], pch = 1, col = "red", cex = 0.2)
```

1.27

1.28

1.29

1.30

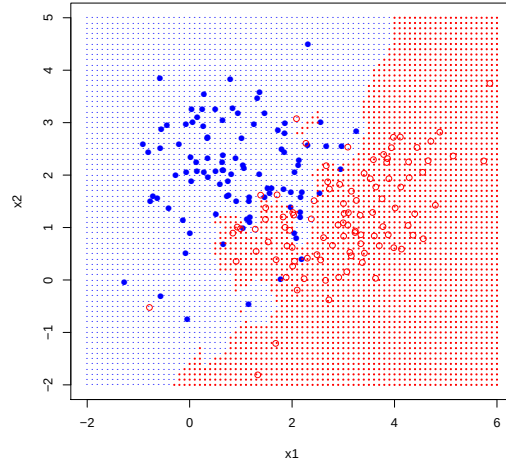


Figure 2: Classification of the points on the grid by using the k -nearest neighbours algorithm, with $k = 3$

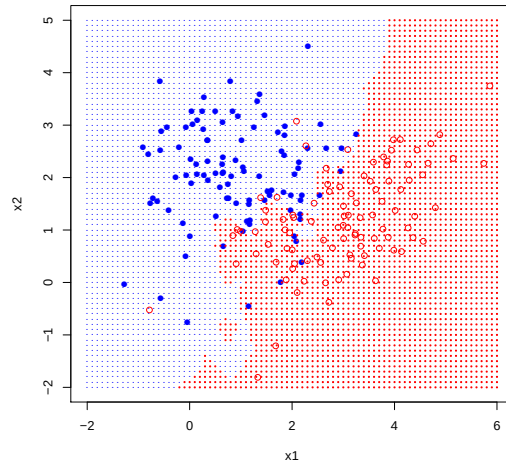


Figure 3: Classification of the points on the grid by using the k -nearest neighbours algorithm, with $k = 5$

8 A Brief History of Statistical Learning

- Legendre Gauss: least squares criterion
 - Galton (1886): linear regression
Goals
 - prediction of a *quantitative* variable Y
 - study of the relationship between Y and the explanatory variables $X_1, \dots, X_k$
 - Fisher (1936) linear discriminant analysis
Goal
 - prediction of a *qualitative* variable Y
-
- (1940-1950) logistic regression
Goals
 - prediction of a *qualitative* variable Y
 - study of the relationship between Y and the explanatory variables $X_1, \dots, X_k$
 - Nelder-Wedderburn (1970-1975) generalized linear models
Goals/Properties
 - prediction of a *quantitative/qualitative* variable Y
 - study of the relationship between Y and the explanatory variables $X_1, \dots, X_k$
 - the authors unified the methodological approach for lm and glm
-
- (1985-20xx) continuous improvement of computational capabilities
 - Breiman, Friedman, Olshen, Stone (1985-1990) classification trees, regression trees, cross-validation procedures
 - Hastie, Tibshirani (1986) generalized additive models
 - (1990 → 20xx) statistical learning & machine learning
 - Hastie, Tibshirani (2001) The Elements of Statistical Learning
reference manual on the fundamentals of *statistical machine learning*
 - James, Witten, Hastie, Tibshirani (2013) An Introduction to Statistical Learning
less theoretical approach, transition of statistical learning from an academic to a mainstream field

1.31

1.32

1.33

9 More on Statistical Learning Goals

We will consider different statistical learning techniques by having regard to their possible applications.

- One must always remind that none of the considered approaches can solve all types of problems.
- Therefore, it is necessary to know the architecture of all techniques, with reference to their potentialities and their weaknesses.

1.34

Let us assume that the theoretical relationship between the (outcome) variable Y and the set of explanatory variables $X_1, X_2, \dots, X_k$ can be described by the relationship

$$Y = f(X_1, X_2, \dots, X_k) + \varepsilon$$

By using statistical learning methods it is possible to obtain an estimate $\hat{f}$ of f . The choice of the most appropriate statistical learning method depends on the analysis goal, that can be classified as *inference*, *prediction* or a combination of the two.

1.35

Inference

If we are studying a metric outcome variable we want to give an answer the following questions.

- Which explanatory variables are associated with the response?
- What is the relationship between the response and each predictor?
 - direct relationships?
 - inverse relationships?
 - is any interaction present?
- Can the relationship between Y and each predictor be adequately summarized using a linear equation, or is the relationship more complicated?

1.36

Prediction

The accuracy of the prediction of Y obtained with the estimate $\hat{f}$ of f depends on two sources of error, which are present in the model

$$Y = f(X_1, X_2, \dots, X_k) + \varepsilon$$

- reducible error: $\hat{f}$ is an approximation of f ; it is possible to reduce this error by using the most appropriate statistical learning method that gives the best approximation of f (we have always to remember that the true/exact form of f is usually unknown);
- irreducible error: this error is given by the presence of the random component ε in the specification of the theoretical model

$$Y = f(X_1, X_2, \dots, X_k) + \varepsilon$$

ε summarizes the contribution of the variables that are not included in the specification f , as well as of the natural variability that characterizes all experiments.

If $\hat{f}$ and $X_1, X_2, \dots, X_k$ are fixed it follows that $E(\varepsilon \hat{f}) = E(\varepsilon f) = 0$, and it can be proven that

1.37

$$\begin{aligned} E \left[\{Y - \hat{Y}\}^2 \right] &= E \left[\{f(X_1, X_2, \dots, X_k) + \varepsilon - \hat{f}(X_1, X_2, \dots, X_k)\}^2 \right] = \\ &= \underbrace{[\hat{f}(X_1, X_2, \dots, X_k) - f(X_1, X_2, \dots, X_k)]^2}_{\text{reducible}} + \underbrace{\text{Var}(\varepsilon)}_{\text{irreducible}} \end{aligned}$$

1.38

Let us classify some problems:

Problem 1

A company is interested in conducting a direct-marketing campaign. The goal is to identify individuals who will respond positively to a mailing, based on observations of demographic variables measured on each individual.

This is an example of modeling for *prediction*.

1.39

Problem 2

Consider the Advertising data illustrated in Figure 2.1 in JWHT. One may be interested in answering questions such as:

- Which media contribute to sales?

- Which media generate the biggest boost in sales? or
- How much increase in sales is associated with a given increase in TV advertising?

This situation falls into the *inference* paradigm.

1.40

Problem 3

1.41

Consider the modelling the brand of a product that a customer might purchase based on variables such as price, store location, discount levels, competition price, and so forth. In this situation one might really be most interested in how each of the individual variables affects the probability of purchase. For instance, what effect will changing the price of a product have on sales?

This is an example of modelling for *inference*.

1.42

Problem 4

In a real estate setting, one may seek to relate values of homes to inputs such as crime rate, zoning, distance from a river, air quality, schools, income level of community, size of houses, and so forth.

If one is interested in how the individual input variables affect the prices – that is, how much extra will a house be worth if it has a view of the river, we are in presence of an *inference* problem.

If one is simply interested in predicting the value of a home given its characteristics – say, is this house under- or over-valued,

we are in presence of a *prediction* problem.

1.43

Remark

Depending on whether our ultimate goal is prediction, inference, or a combination of the two, different methods for estimating f may be appropriate.

Remark

Statistical learning procedures are aimed to obtain an estimate $\hat{f}$ of the relationship f and not to search for the estimate $\hat{f}$ that best fits the data.

By using the function $\hat{f}$ that best fits the data, one could encounter a so-called *overfitting* problem: a too much flexible model (that is a model that has a quite good fit to the data) can describe some behaviour that is present in data, but that is due to the presence of the error ε and not to the structure f .

In this case the estimate $\hat{f}$ will follow the error or *noise* too closely.

1.44

10 Estimation of $f(\cdot)$

Parametric methods

1. *First*, we make an *assumption* about the functional form of $f(\cdot)$.
For example a linear model specification

$$f(X_1, X_2, \dots, X_k) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

2. We need a procedure that uses the training data to *fit* or *train* the model; that is a procedure that estimates the model parameters (we can use OLS to estimate the linear model).

Pros

A simple specification of $f(\cdot)$ can be given, and each parameter can be interpreted.

Drawbacks

If we choose a *specification* f very far from the *true* f our estimate will be poor, and predictions of Y will not be reliable.

The problem can be addressed by choosing a model with a higher level of *flexibility*; e.g., by including non-linear relationships, interaction effects, etc. in the linear model specification;

in this way we are increasing the number of parameters in the model.

One has also to take into account that a too much flexible model can lead to *overfitting* problems.

Non-parametric methods

Non-parametric methods do not make explicit assumptions about the functional form of $f(\cdot)$. Instead they seek an estimate of f that gets as closed to the data points without being too rough or wiggly.

One example is given by simple moving averages that are used to estimate the trend of a seasonal time series with period k

- with moving averages of order k one finds an estimate of the trend that removes only the seasonal component
- with moving averages of order $2k$ the estimate of the trend is smoother; part of the noise is removed or, possibly, the time series is filtered from some cyclic component (as it happens, for example, with moving averages of order 2 and 4 applied to a bi-annual time series, that is characterized also by a 2 years cycle)
- ...

Pros

No assumption about the form of f is made. Non-parametric methods have the potential to accurately fit a wider range of possible shapes for f .

Drawback

A very large number of observations (far more than is typically needed for a parametric approach) is required in order to obtain an accurate estimate for f .

Consider the example of the loss of information that characterizes the moving averages estimate of the trend of a time series.

Example 4. We consider six non-parametric estimates of the relationship between the wage and the number of years of experience for the data set of Belgium wages.

- the regression function that relates each value of the explanatory variable, $EXPER$, with the conditional average of the outcome, $WAGE$,

$$i \rightarrow E(WAGE|EXPER = i), \quad i = 0, 1, 2, \dots, \max(EXPER)$$

red curve in the left side of Figure 4

- a first smoothed version obtained as

$$i \rightarrow E(WAGE|i-1 \leq EXPER \leq i+1), \quad i = 1, 2, \dots, \max(EXPER) - 1$$

blue curve in the left side of Figure 4

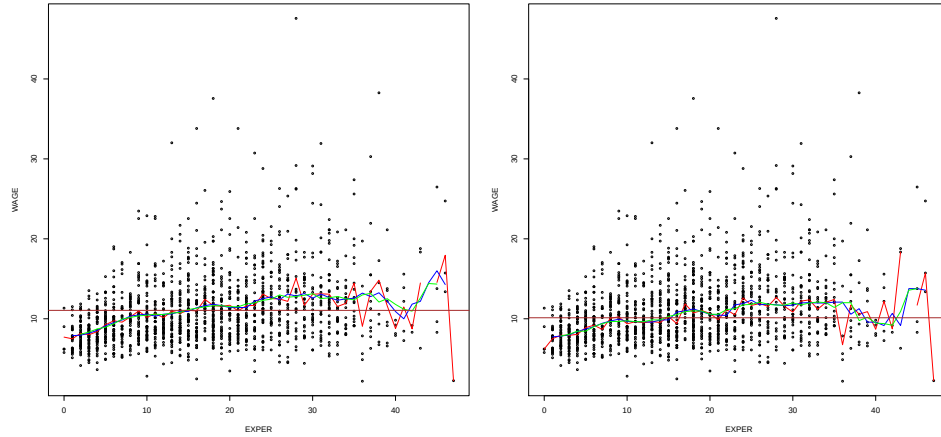


Figure 4: Estimates of the mean and median level of the wage conditional on the number of years of experience

- a second smoothed version obtained as

$$i \rightarrow E(WAGE|i-2 \leq EXPER \leq i+2), \quad i = 2, \dots, \max(EXPER) - 2$$

green curve in the left side of Figure 4

1.50

- the function that relates each value of the explanatory variable, $EXPER$, with the conditional median (percentage point of order 0.5) of the outcome, $WAGE$,

1.51

$$i \rightarrow (WAGE|EXPER = i)_{0.5}, \quad i = 0, 1, 2, \dots, \max(EXPER)$$

red curve in the right side of Figure 4

- a first smoothed version obtained as

$$i \rightarrow (WAGE|i-1 \leq EXPER \leq i+1)_{0.5}, \quad i = 1, 2, \dots, \max(EXPER) - 1$$

blue curve in the right side of Figure 4

- a second smoothed version obtained as

$$i \rightarrow (WAGE|i-2 \leq EXPER \leq i+2)_{0.5}, \quad i = 2, \dots, \max(EXPER) - 2$$

green curve in the right side of Figure 4

General mean and median levels are also reported as brown lines.

1.52

Figure 4 can be reproduced with the code

```
layout(matrix(1:2, 1, 2))
with(indwages, plot(WAGE ~ EXPER, cex = 0.5))
set <- with(indwages, min(EXPER) : max(EXPER))
a <- with(indwages, sapply(set, function(i) mean(WAGE[EXPER ==
i])))
lines(set, a, col = "red", lwd = 2)
set <- with(indwages, (min(EXPER) + 1) : (max(EXPER) - 1))
a <- with(indwages, sapply(set, function(i) mean(WAGE[(EXPER >=
i - 1) & (EXPER <= i + 1)])))
```

```

lines(set, a, col = "blue", lwd = 2)
set <- with(indwages, (min(EXPER) + 2):(max(EXPER) - 2))
a <- with(indwages, sapply(set, function(i) mean(WAGE[(EXPER >=
  i - 2) & (EXPER <= i + 2)])))
lines(set, a, col = "green", lwd = 2)
with(indwages, abline(h = mean(WAGE), col = "brown"))

```

1.53

```

with(indwages, plot(WAGE ~ EXPER, cex = 0.5))
set <- with(indwages, min(EXPER):max(EXPER))
a <- with(indwages, sapply(set, function(i) median(WAGE[EXPER ==
  i])))
lines(set, a, col = "red", lwd = 2)
set <- with(indwages, (min(EXPER) + 1):(max(EXPER) - 1))
a <- with(indwages, sapply(set, function(i) median(WAGE[(EXPER >=
  i - 1) & (EXPER <= i + 1)])))
lines(set, a, col = "blue", lwd = 2)
set <- with(indwages, (min(EXPER) + 2):(max(EXPER) - 2))
a <- with(indwages, sapply(set, function(i) median(WAGE[(EXPER >=
  i - 2) & (EXPER <= i + 2)])))
lines(set, a, col = "green", lwd = 2)
with(indwages, abline(h = median(WAGE), col = "brown"))

```

1.54

11 The Trade-Off Between Prediction Accuracy and Model Interpretability

Figure extracted from James, Witten, Hastie, Tibshirani (2013) An Introduction to Statistical Learning, Springer

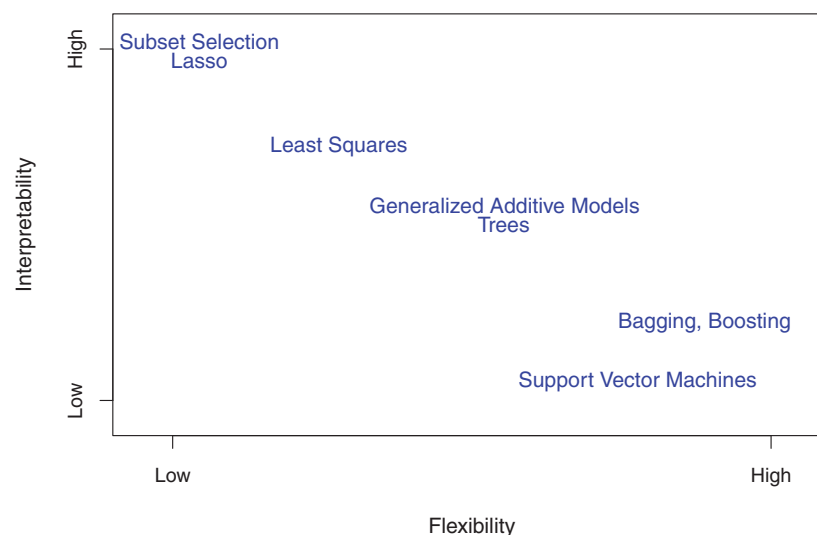


FIGURE 2.7. A representation of the tradeoff between flexibility and interpretability, using different statistical learning methods. In general, as the flexibility of a method increases, its interpretability decreases.

1.55